

CS 30700: Project Charter

Team Number : 25

Members : Arya Shinde, Digvijay Pawar, Mridhula Srinivasan, Pradyumna Danigond, Sayam Kanwar

Project Title : *SmartShopper*

Problem Statement

Large supermarkets including Walmart and Target serve the general public with all kinds of different goods serving the broad to very specific needs, however, a store of this size becomes incredibly hard to navigate for a simple grocery run. While shopping at these supermarkets, people often end up investing more time and money into the process. Using SmartShopper, people will easily be able to optimize their shopping process by following a calculative store path suited for their grocery list. The app would also be able to suggest particular recipes of the person's interest and preferences based on the items in their grocery inventory. The existing solutions include the aisle locations of the items in Walmart and Target apps but there exists no such integrated solution that can help people save their time, money, and effort.

Project Objectives

An application that has the following features:

1. Route optimization in supermarkets based on shopping lists to reduce the amount of time spent in searching for products and avoiding distractions
 - The application displays the next closest item on the list and how to get there from the current position beginning from time of entry in the store, till the last item is picked up.
 - It will also differentiate the list based on availability of the items added to the list in the store the consumer chooses to use. It can do so by not displaying those items when showing you the next closest items or by displaying close alternatives in the store the consumer is in.
2. Price comparison of products at different stores to help users choose where they want to buy from
 - Aggregates prices of different stores in one platform to show you the cheapest offering of a particular product or equivalent products when the product is not available at another store.
 - Display reviews for competing types of items if available to help consumers make a more informed choice.
3. Suggesting recipe ideas based on products shopped and already available at home, to maximize the use of products stocked at home and reduce waste
 - Consumers can choose to add items already at home along with the items in the shopping list, or simply add ingredients currently available to get recipe ideas for those ingredients.
 - Display suggested recipes, and if the user is interested, the ingredients get added to the shopping list, if not already present at home.
4. Reminding users to restock their groceries by keeping a track of ingredients at home, and last time they were bought.

- Can be used as a way to track the inventory of grocery items at home. (with the quantity being an editable option, as per the user's wishes)
- Checks last instance they were added to the shopping list and if time exceeds the number of days the item was safely consumable for, it is suggested to add to the shopping list.

Stakeholders

Product Developers: Arya Shinde, Digvijay Pawar, Mridhula Srinivasan, Pradyumna Danigond, Sayam Kanwar

Users: Customers of supermarkets (general public), employees working in supermarkets, and Suppliers/Vendors (supermarkets themselves)

Project Owners: Arya Shinde, Digvijay Pawar, Mridhula Srinivasan, Pradyumna Danigond, Sayam Kanwar

Project Deliverables

1. Flutter-based mobile application that allows the user to make a list of groceries that they need to purchase and an existing list of ingredients already available
2. Python Flask-based backend API to generate the shortest route to access grocery items and provide recipe suggestions
 - a. For gathering the data required (item's aisle locations and ingredients used in different recipes), we will be scraping the web, particularly Walmart.com.
3. Firebase database for storing the user grocery lists and profiles